Validity Analysis: VISDOMBENCH

Target context: Bangladesh Atomic Energy Commission — Internal Document Repository QA

Overall risk: **HIGH**

Deployment Context

Use case: This is valid for government scientific facilities where you need to prepare a repository QA system for internal documents separated from the internet because of document sensitivity. Now the system is ideal for any country/organization based government facilities but you need to adapt the model so that it understands the documents from target domain.

Target population: Bangladesh Atomic Energy Commission. The language contains literature from the nuclear energy science domain in english and german languages. As well as, a separate official document repository that contains PDF documents in Bengali and English.

Dimension Scores

Dimension	Score	Rating	Confidence
Input Ontology ✓	2	Significant gaps	high
Input Content ■	1	Serious concern	high
Input Form ■	1	Serious concern	high
Output Ontology ✓	2	Significant gaps	medium
Output Content ■	1	Serious concern	high
Output Form	2	Significant gaps	high
Average	**1.5**		

■ = highest concern ✓ = strongest dimension

Dimension Key

Abbr.	Dimension	Definition
IO	Input Ontology	Whether the benchmark's test case categories cover the query types expected in deployment.
IC	Input Content	Whether individual datapoint content is culturally and contextually appropriate for the target region.
IF	Input Form	Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.
OO	Output Ontology	Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.
OC	Output Content	Whether ground-truth labels align with the judgments of regional stakeholders.
OF	Output Form	Whether the expected output modality matches regional deployment needs and accessibility.

Overall Summary

VisDoMBench provides a methodologically sound reference architecture for multi-document multimodal PDF QA with evidence attribution — structurally analogous to BAEC's internal repository QA use case. However, every content-level and language-level dimension exhibits severe or complete mismatch with the BAEC deployment. The benchmark is English-only, drawn exclusively from CS/ML ArXiv papers and academic slide decks, annotated by a single CS graduate

student, scored with language-sensitive Word Overlap F1, and dependent on cloud LLM APIs incompatible with BAEC's air-gapped environment. BAEC's deployment requires Bengali-script OCR (where PyTesseract systematically underperforms), multilingual retrieval across Bengali/English/German/Russian corpora (where SoTA cross-lingual Recall@1 remains below 30% on Bengali), nuclear-domain terminological coverage (a documented capability gap even for SoTA LLMs), and expert annotation by nuclear scientists and BAERA regulatory officers. The only transferable value is architectural: the triple (q, D, â) formulation, dual textual + visual RAG pipelines, evidence curation with attribution, and provenance-aware retrieval metrics (ANLCS, document-identification rate) are patterns worth replicating in a BAEC-specific benchmark that would need to be built essentially from scratch.

Practical Guidance

What This Benchmark Measures

VisDoMBench measures the ability of multimodal RAG systems to answer questions over multi-document English-language academic PDF collections (CS/ML ArXiv papers, scientific chart datasets, academic slide decks) with evidence attribution across pages. Its strongest dimensions for BAEC are the Input Form architectural pattern (PDF ingestion, text+visual parallel retrieval, metadata preservation) and the Output Form provenance-attribution metrics (ANLCS, document-identification rate). It can serve as a methodological reference for multi-document QA system design.

Construct Depth

The benchmark probes multi-document retrieval, cross-page reasoning, and evidence attribution deeply within English academic CS/ML content, with multiple reasoning categories (single-hop, multi-hop, numerical) and robust ablations (early vs. late fusion, page-count scaling). However, it provides zero construct depth for nuclear-domain knowledge, Bengali or multilingual document understanding, regulatory/compliance QA, government administrative queries, or safety-critical output reliability. Separate benchmarks (NuclearQA, NuclearQAv2, nuclear regulatory FSAR QA, KhatianDoc for Bengali legal documents) exist specifically to measure these constructs.

What Else You Need

A BAEC-valid evaluation requires: (a) a Bengali administrative document QA set with human-verified ground truth by Bengali-literate BAEC staff (addressing Input Content, Input Form, Output Content gaps); (b) a nuclear-domain QA set drawing on IAEA/BAERA standards and Rosatom VVER-1200 documentation, annotated by nuclear engineers (addressing Input Ontology, Input Content, Output Content gaps); (c) evaluation of multilingual embeddings (mE5-large-instruct, BGE-M3) and bbOCR for Bengali on actual BAEC corpora (addressing Input Form gaps); (d) replacement of Word Overlap F1 with script-aware and faithfulness-oriented metrics (e.g., RAGAS Faithfulness as used in the nuclear regulatory QA benchmark) for multilingual safety-critical scoring (addressing Output Form gaps); (e) empirical measurement of hallucination rates on nuclear safety threshold queries. Existing benchmarks NuclearQA [WEB-20], NuclearQAv2 [WEB-21], the nuclear regulatory FSAR benchmark [WEB-22], and KhatianDoc [WEB-11] should be used as external capability probes alongside any BAEC-specific evaluation.

ID	Evidence
WEB-11	Bengali document QA requires specialized evaluation; current benchmarks use human-verified ground truth rather than lexical overlap (https://arxiv.org/pdf/2609.03597)
WEB-20	NuclearQA was expert-authored (100 questions) — nuclear-domain correctness requires expert annotation (https://arxiv.org/pdf/2310.10920)
WEB-21	https://arxiv.org/html/2606.27047
WEB-22	Nuclear regulatory QA benchmarks use faithfulness measures (RAGAS Faithfulness 0.93) rather than Word Overlap F1 to assess safety-critical correctness (https://arxiv.org/html/2606.29399v1)

Input Ontology (IO)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: *VISDOMBENCH* | ***Context:*** *Bangladesh Atomic Energy Commission — Internal Document Repository QA*

Input Ontology definition: Whether the benchmark's test case categories cover the query types expected in deployment.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

VisDoMBench's task ontology — multi-document QA over PDFs with evidence attribution and single-hop, multi-hop, and numerical reasoning [Q104, Q19, Q20] — is structurally analogous to BAEC's repository QA use case. However, the category coverage is entirely drawn from academic CS/ML publications (ArXiv papers, chart QA, slide decks) [Q25, Q26, Q100], with no representation of regulatory lookups, procedural retrieval, administrative queries, or nuclear safety threshold queries that constitute BAEC's expected query distribution. Nuclear-domain benchmarks have been constructed separately (NuclearQA, NuclearQAv2, and a NuScale FSAR multi-hop benchmark) precisely because such categories are absent from general multi-document QA benchmarks [WEB-20, WEB-21, WEB-22], and Bengali government administrative QA has no benchmark coverage at all [WEB-24, WEB-25]. This represents significant construct underrepresentation for the BAEC deployment.

ID	Evidence
Q19	“Given a query q, we have a collection of M documents D = {d1, d2, . . . , dM}, wherein each document di may consist of a set of Ni pages represented by P i = {pi 1, pi 2, . . . , pi Ni}. We aim to generate text aˆ for each query q that accurately answers the user query.” (p.3)
Q20	“The answer generation relies on retrieving relevant evidence context from one or more documents. Each query q may require information spread across different pages from one or more of the associated documents in D.” (p.3)
Q25	“Our corpus comprises test/eval sets sourced from several established datasets. We incorporated the PaperTab and FeTaTab splits from the UDA Benchmark(Hui et al., 2024), which in turn sourced these datasets from QASPER(Dasigi et al., 2021) and FeTaQA(Nan et al., 2022), respectively.” (p.3)
Q26	“For chart-based question-answering samples, we drew from SciGraphQA (Li and Tajbakhsh, 2023), which is multi-turn QA dataset on charts from scientific papers, and SPIQA(Pramanick et al., 2024), a chart and table QA dataset system sourced from (Dasigi et al., 2021). Additionally, we included SlideVQA (Tanaka et al., 2023), a multi-image, multi-hop QA dataset centered on presentation slide decks.” (p.3)
Q100	“The dataset was constructed from 290,000 Computer Science and Machine Learning papers published on ArXiv between 2010 and 2020, with the help of Palm-2 generating 295,000 samples of open-vocabulary multi-turn question-answer dialogues about the graphs.” (p.14)
Q104	“This dataset requires complex reasoning skills, including single-hop, multi-hop, and numerical reasoning.” (p.14)
WEB-20	NuclearQA was expert-authored (100 questions) — nuclear-domain correctness requires expert annotation (https://arxiv.org/pdf/2310.10920)
WEB-21	https://arxiv.org/html/2606.27047
WEB-22	Nuclear regulatory QA benchmarks use faithfulness measures (RAGAS Faithfulness 0.93) rather than Word Overlap F1 to assess safety-critical correctness (https://arxiv.org/html/2606.29399v1)
WEB-24	MMDocIR includes a Government domain (text-centric, 88.2% text questions) distinct from academic QA (https://aclanthology.org/2025.emnlp-main.1576.pdf)
WEB-25	SDS KoPub VDR — Korean public document retrieval benchmark, closest analog to Bengali government administrative QA (which has no benchmark coverage) (https://arxiv.org/pdf/2511.04910)

Strengths

- The abstract task formulation — triple (q, D, â) with evidence attribution across multiple pages [Q19, Q22] — mirrors BAEC's repository QA architecture

• Inclusion of single-hop, multi-hop, and numerical reasoning categories [Q104] covers the structural reasoning types BAEC users would employ (e.g., cross-document safety comparison, parameter lookup)

• Explicit treatment of in-domain distractor documents [Q106, Q107] simulates realistic retrieval difficulty analogous to a multi-institute BAEC corpus

ID	Evidence
Q19	“Given a query q, we have a collection of M documents D = {d1, d2, . . . , dM}, wherein each document di may consist of a set of Ni pages represented by P i = {pi 1, pi 2, . . . , pi Ni}. We aim to generate text aˆ for each query q that accurately answers the user query.” (p.3)
Q22	“Every data point in VisDoMBench can be expressed as triple (q, D, aˆ), where a question q is posed to a set of documents D, with groundtruth answer aˆ.” (p.3)
Q104	“This dataset requires complex reasoning skills, including single-hop, multi-hop, and numerical reasoning.” (p.14)
Q106	“Distracting documents are introduced as additional, irrelevant documents within the retrieval set to simulate real-world scenarios where the task is to find the most relevant context among multiple documents.” (p.14)
Q107	“These distracting documents are selected randomly from the in-domain documents of a given dataset, ensuring that they are contextually similar but not directly relevant to the query.” (p.14)

Checklist

Checklist item definitions:

ID	Assessment Question
IO-1	Identify test case categories required for regional deployment
IO-2	Check if taxonomy omits regionally relevant categories
IO-3	Check if taxonomy includes irrelevant categories
IO-4	Document category gaps harming content validity

Checklist responses:

IO-1: BAEC-relevant categories include: safety regulation and threshold lookups (IAEA/BAERA), technical procedure retrieval (reactor operations), cross-document regulatory synthesis, Bengali administrative document retrieval, and numerical nuclear parameter lookups (cross-sections, isotope inventories). None of these categories appear in VisDoMBench's taxonomy, which is drawn exclusively from CS/ML academic datasets [Q25, Q26].

IO-2: Yes — the taxonomy omits regulatory QA, procedural manual retrieval, administrative document QA, and any government or nuclear-domain query category. Nuclear-domain benchmarks exist as separate efforts [WEB-20, WEB-21, WEB-22] specifically because such categories are absent from general-purpose benchmarks like VisDoMBench.

IO-3: The taxonomy is not strictly irrelevant — academic chart/table/slide QA is structurally analogous to the modalities BAEC encounters — but the specific instantiations (ML research charts, academic slide decks) are not representative of nuclear engineering diagrams or Bengali administrative forms.

IO-4: Category gaps that harm content validity for BAEC: (a) regulatory/compliance QA, (b) procedural/safety retrieval, (c) Bengali administrative queries, (d) multilingual cross-lingual queries (Bengali query over English/German/Russian document). VisDoMBench provides no coverage of these [Q23, Q24].

ID	Evidence
Q23	“We re-purpose five existing document-QA datasets to form our benchmark.” (p.3)
Q24	“In the curation of document question-answering datasets, we adhered to the following criteria: (1) the inclusion of visually rich content, encompassing tables, charts, and presentation slides; (2) the utilization of publicly accessible source documents; and (3) the presence of grounded evidence.” (p.3)
Q25	“Our corpus comprises test/eval sets sourced from several established datasets. We incorporated the PaperTab and FeTaTab splits from the UDA Benchmark(Hui et al., 2024), which in turn sourced these datasets from QASPER(Dasigi et al., 2021) and FeTaQA(Nan et al., 2022), respectively.” (p.3)

Q26	“For chart-based question-answering samples, we drew from SciGraphQA (Li and Tajbakhsh, 2023), which is multi-turn QA dataset on charts from scientific papers, and SPIQA(Pramanick et al., 2024), a chart and table QA dataset system sourced from (Dasigi et al., 2021). Additionally, we included SlideVQA (Tanaka et al., 2023), a multi-image, multi-hop QA dataset centered on presentation slide decks.” (p.3)
WEB-20	NuclearQA was expert-authored (100 questions) — nuclear-domain correctness requires expert annotation (https://arxiv.org/pdf/2310.10920)
WEB-21	https://arxiv.org/html/2606.27047
WEB-22	Nuclear regulatory QA benchmarks use faithfulness measures (RAGAS Faithfulness 0.93) rather than Word Overlap F1 to assess safety-critical correctness (https://arxiv.org/html/2606.29399v1)
WEB-24	MMDocIR includes a Government domain (text-centric, 88.2% text questions) distinct from academic QA (https://aclanthology.org/2025.emnlp-main.1576.pdf)
WEB-25	SDS KoPub VDR — Korean public document retrieval benchmark, closest analog to Bengali government administrative QA (which has no benchmark coverage) (https://arxiv.org/pdf/2511.04910)

Information Gaps

- Actual query-type distribution expected at BAEC is inferred from documented facility structure, not from an elicited user-query log

Requires Expert Verification

- BAEC stakeholder confirmation of the operational query type distribution (regulatory lookup vs. procedural retrieval vs. administrative queries) and their relative frequencies

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Remediation

Gap 1: Taxonomy omits regulatory lookup, procedural retrieval, administrative query, and multilingual cross-lingual query categories essential to BAEC

Recommendation: Construct a BAEC-specific query taxonomy from stakeholder elicitation covering IAEA/BAERA safety threshold lookups, reactor procedural retrieval, cross-document regulatory synthesis, Bengali administrative queries, and cross-lingual retrieval scenarios; use NuScale FSAR nuclear regulatory benchmark [WEB-22] as a structural template.

ID	Evidence
WEB-22	Nuclear regulatory QA benchmarks use faithfulness measures (RAGAS Faithfulness 0.93) rather than Word Overlap F1 to assess safety-critical correctness (https://arxiv.org/html/2606.29399v1)

Input Content (IC)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: *VISDOMBENCH* | ***Context:*** *Bangladesh Atomic Energy Commission — Internal Document Repository QA*

Input Content definition: Whether individual datapoint content is culturally and contextually appropriate for the target region.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

All VisDoMBench input content is drawn from publicly available English-language academic sources — 290,000 ArXiv CS/ML papers [Q100], scientific chart QA, and academic slide decks [Q103] — with query augmentation performed by GPT-4o on CS/ML content [Q40, Q41]. For BAEC, the input content is a complete domain and linguistic mismatch: the target corpus includes Bengali official government documents, English/German nuclear engineering literature, and likely Russian-language Rosatom VVER-1200 documentation [WEB-3, WEB-4, WEB-19]. None of the vocabulary, document structures, citation conventions, or subject matter of nuclear science, government administration, or non-English content is represented. Nuclear-domain QA has been shown to be a distinct capability gap even for SoTA LLMs [WEB-20], and Bengali legal document QA fails near-completely on current multimodal models [WEB-11].

ID	Evidence
Q40	“To address the challenge of ambiguous questions in datasets such as SciGraphQA, and PaperTab, we implement a query augmentation procedure to create a one-to-one mapping between a given question and the document(s) that exclusively answer it.” (p.4)
Q41	“Given an original question and the document containing answer, we utilize GPT-4o to generate more specific variations of the question, ensuring that the generated question can only be answered by the corresponding document.” (p.4)
Q100	“The dataset was constructed from 290,000 Computer Science and Machine Learning papers published on ArXiv between 2010 and 2020, with the help of Palm-2 generating 295,000 samples of open-vocabulary multi-turn question-answer dialogues about the graphs.” (p.14)
Q103	“SlideVQA (Tanaka et al., 2023) is a multi-image document VQA dataset that contains over 2,600 slide decks, comprising more than 52,000 slide images and 14,500 questions regarding the slide content.” (p.14)
WEB-3	Rooppur NPP constructed under Rosatom contract; Russian-language technical documentation highly likely in BAEC corpus (https://world-nuclear.org/information-library/country-profiles/countries-a-f/bangladesh)
WEB-4	Russian-language signage and workers at Rooppur site corroborate Russian document presence (https://fvbb.com/ns/from-sleepy-village-to-nuclear-hub-rooppur/323313)
WEB-11	Bengali document QA requires specialized evaluation; current benchmarks use human-verified ground truth rather than lexical overlap (https://arxiv.org/pdf/2609.03597)
WEB-19	Rooppur NPP uses VVER-1200 reactors with Russian design specifications and construction standards (https://www.gem.wiki/Rooppur_nuclear_power_plant)
WEB-20	NuclearQA was expert-authored (100 questions) — nuclear-domain correctness requires expert annotation (https://arxiv.org/pdf/2310.10920)

Strengths

- The augmentation methodology (specifying queries using document metadata, titles, abstracts, captions) [Q110] is methodologically transferable to a BAEC corpus if re-executed with domain content

ID	Evidence
Q110	“We addressed the challenge of trivial or underspecified queries in some datasets by augmenting the queries using ChatGPT4o and relevant context, including the title and abstract of the research paper, the relevant figure's caption, and other available metadata.” (p.15)

Checklist

Checklist item definitions:

ID	Assessment Question
IC-1	Do inputs require region-specific cultural/geographic/dialectal knowledge?
IC-2	Does culturally sensitive content align with target culture?
IC-3	Flag inputs assuming Western-specific knowledge
IC-4	Regional annotator recruitment for culturally sensitive instances
IC-5	Document content issues harming content validity

Checklist responses:

IC-1: Yes — BAEC queries require Bengali-language diglossic formal register knowledge, nuclear reactor physics terminology (IAEA/ISO/DIN conventions), and potentially Russian/Cyrillic Rosatom documentation familiarity. VisDoMBench content is English-only and CS/ML academic [Q23, Q85, Q100].

IC-2: VisDoMBench content is not culturally sensitive in a way that offends Bangladeshi norms, but it is culturally irrelevant: Western academic CS/ML content does not reflect the vocabulary, institutional conventions, or document types of BAEC's repository [Q100].

IC-3: Yes — the content presupposes familiarity with ArXiv CS/ML research conventions, Western academic citation norms, and English scientific prose [Q100]. None of this transfers to Bengali administrative documents or German nuclear safety reports.

IC-4: Regional annotator recruitment (Bengali-proficient, nuclear-domain-literate) would be required to construct a BAEC-representative content set; the benchmark provides no such content and no methodology for sourcing it.

IC-5: Content issues harming content validity: (a) zero Bengali content; (b) zero German or Russian content; (c) zero nuclear science domain content; (d) zero government administrative document content. Each of these corresponds to a documented capability gap in current models [WEB-11, WEB-20, WEB-22].

ID	Evidence
Q23	“We re-purpose five existing document-QA datasets to form our benchmark.” (p.3)
Q43	“Once the augmented queries are generated, a human annotator reviews them using a predefined rubric. The rubric guides the annotator to either select one of the five generated questions, retain the original question, or mark all questions (synthetic and actual) as ambiguous, in which case, the data point is discarded.” (p.5)
Q85	“We use publicly available datasets in this research.” (p.9)
Q100	“The dataset was constructed from 290,000 Computer Science and Machine Learning papers published on ArXiv between 2010 and 2020, with the help of Palm-2 generating 295,000 samples of open-vocabulary multi-turn question-answer dialogues about the graphs.” (p.14)
WEB-3	Rooppur NPP constructed under Rosatom contract; Russian-language technical documentation highly likely in BAEC corpus (https://world-nuclear.org/information-library/country-profiles/countries-a-f/bangladesh)
WEB-11	Bengali document QA requires specialized evaluation; current benchmarks use human-verified ground truth rather than lexical overlap (https://arxiv.org/pdf/2609.03597)
WEB-20	NuclearQA was expert-authored (100 questions) — nuclear-domain correctness requires expert annotation (https://arxiv.org/pdf/2310.10920)
WEB-22	Nuclear regulatory QA benchmarks use faithfulness measures (RAGAS Faithfulness 0.93) rather than Word Overlap F1 to assess safety-critical correctness (https://arxiv.org/html/2606.29399v1)

Information Gaps

- Exact proportion of Bengali vs. English vs. German vs. Russian material in BAEC's actual repository is not publicly documented

Requires Expert Verification

- BAEC stakeholders would need to confirm language mix proportions, extent of Russian-language Rosatom documentation, and whether German nuclear literature is substantially represented

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Remediation

Gap 1: Zero coverage of Bengali, German, Russian, nuclear science, or government administrative content; benchmark is entirely CS/ML ArXiv academic material

Recommendation: Curate a BAEC-representative content set drawing on IAEA technical documents, BAERA regulatory filings, Rosatom VVER-1200 engineering documentation (Russian), German nuclear safety literature (if confirmed present), and Bengali administrative PDFs; apply VisDoMBench's query augmentation methodology [Q110] but with domain-expert review.

ID	Evidence
Q110	“We addressed the challenge of trivial or underspecified queries in some datasets by augmenting the queries using ChatGPT4o and relevant context, including the title and abstract of the research paper, the relevant figure's caption, and other available metadata.” (p.15)

Input Form (IF)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: *VISDOMBENCH* | ***Context:*** *Bangladesh Atomic Energy Commission — Internal Document Repository QA*

Input Form definition: Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

VisDoMBench ingests PDF documents and extracts text via PyTesseract OCR [Q55], chunks at 3000 characters with 10% overlap [Q56], and uses English-centric retrieval models BM25, MPNet, MiniLM, BGE-1.5 [Q54] plus visual retrievers ColPali/ColQwen2 [Q57]. PDF input is structurally compatible with BAEC. However, the OCR and retrieval pipeline is invalid for Bengali: PyTesseract systematically under-recalls Bengali documents and bbOCR — the SoTA open-source Bengali OCR — achieves only ~75% recognition accuracy on BCD3 [WEB-5, WEB-6]. English-centric embeddings are inadequate for Bengali; multilingual alternatives (mE5-large-instruct, BGE-M3, LaBSE) exist but show substantial cross-lingual degradation — Recall@1 of 27.6%, 21.7%, 15.9% respectively on Bengali [WEB-8, WEB-9]. The benchmark explicitly uses proprietary cloud LLM APIs (GPT-4o, Gemini-1.5-Flash) [Q62] which are incompatible with BAEC's air-gapped deployment constraint. German compound-noun tokenization and Russian/Cyrillic handling are not addressed at all.

ID	Evidence
Q54	“We use popular text based retrieval models: BM25 (Robertson et al., 1995) a statistical baseline, and , MPNet (Song et al., 2020), MiniLM (Wang et al., 2020), and BGE-1.5 (Xiao et al., 2023), which represent SoTA dense retrieval baselines.” (p.6)
Q55	“Text extraction from PDF documents is performed using PyTesseract.” (p.6)
Q56	“The extracted text is then segmented into 3000-character chunks using the recursive-split method (Sarmah et al., 2023), with a 10% overlap to mitigate information loss.” (p.6)
Q57	“For visual retrieval, we utilize recent advances late interaction based multi-vector retrieval models built on top of LLMs (Faysse et al., 2024), namely ColPali and ColQwen2, which have PaliGemma (Beyer et al., 2024) and Qwen2 (Yang et al., 2024) as their base LLMs.” (p.6)
Q62	“We include two off-the-shelf models Gemini-1.5-Flash (Reid et al., 2024), and ChatGPT-4o (OpenAI, 2024), as well as Qwen2-VL-7B-Instruct (Yang et al., 2024), an open-source LLM with visual and long context capabilities.” (p.6)
WEB-5	bbOCR outperforms Tesseract on Bengali across all measured domains; Tesseract recall systematically poor on Bengali (https://arxiv.org/pdf/2308.10647)
WEB-6	APSIS-Net recognition in bbOCR achieves ~75% accuracy on BCD3 — the SoTA Bengali OCR ceiling (https://www.researchgate.net/publication/373263373_bbOCR_An_Open-source_Multi-domain_OCR_Pipeline_for_Bengali_Documents)
WEB-8	Cross-lingual Bengali retrieval Recall@1: mE5-large-instruct 27.6%, BGE-M3 21.7%, LaBSE 15.9% — all substantially lower than monolingual settings (https://arxiv.org/pdf/2601.10205)
WEB-9	mE5-large-instruct is leading multilingual embedding but still exhibits ~25% relative cross-lingual performance drop (https://arxiv.org/pdf/2402.05672)

Strengths

- PDF-as-input format [Q7, Q8] matches BAEC's repository medium
- Preservation of source document and page metadata for traceability [Q49] aligns with BAEC's provenance-attribution requirements
- Dual textual + visual retrieval architecture [Q47] is structurally appropriate for nuclear engineering diagrams and administrative documents containing signatures/stamps

ID	Evidence

Q7	“These documents often contain a rich blend of textual, visual, and tabular data, making them a unique challenge for information retrieval systems.” (p.1)
Q8	“Unlike structured formats like databases, PDFs are inherently unstructured, with diverse layouts combining paragraphs, images, charts, and tables.” (p.1)
Q47	“VisDoMRAG (Fig 2) is a multimodal RAG approach for visually rich document QA consisting of two steps: (i) parallel evidence-driven unimodal (vision and textual) RAG pipelines, and (ii) Modality Fusion, which imposes consistency constraints to combine unimodal reasoning chains and arrive at a final answer.” (p.5)
Q49	“Metadata indicating the source document and page number is preserved to facilitate traceability.” (p.5)

Checklist

Checklist item definitions:

ID	Assessment Question
IF-1	Compare signal distributions between source and target
IF-2	Check if regional infrastructure supports data capture specs
IF-3	Identify domain-specific form differences
IF-4	Document form mismatches harming external validity

Checklist responses:

IF-1: Signal distributions differ substantially: VisDoMBench PDFs are born-digital English academic papers from ArXiv [Q35] and English slide decks [Q103]; BAEC inputs include Bengali-script scanned administrative PDFs with signatures/stamps, German technical documents with umlauts and compound nouns, and likely Cyrillic Rosatom documents. OCR and tokenization pipelines designed for the former fail on the latter [WEB-5, WEB-11].

IF-2: BAEC is fully air-gapped — proprietary cloud LLM endpoints (GPT-4o, Gemini-1.5-Flash) used in the primary benchmark configuration [Q62] cannot be deployed. On-premise compute capacity for open-source alternatives (Qwen2-VL-7B) is not publicly documented.

IF-3: Domain-specific form differences: (a) Bengali script OCR is a documented weak point of PyTesseract [WEB-5]; (b) scanned-document artifacts (stamps, signatures) are not addressed by the benchmark; (c) German compound-noun tokenization is not evaluated; (d) Cyrillic handling is entirely absent from the benchmark [Q55, Q56].

IF-4: Form mismatches harming external validity: (a) PyTesseract systematically under-performs on Bengali [WEB-5]; (b) English-centric embeddings fail cross-lingually on Bengali [WEB-8]; (c) cloud LLM dependence incompatible with air-gap; (d) Word Overlap F1 chunk-boundary dependence on Latin-tokenization assumptions.

ID	Evidence
Q35	“SciGraphQA and SPIQA contain questions specific to charts or tables extracted from scientific papers. We use the ArXiv API to extract full document PDFs.” (p.4)
Q54	“We use popular text based retrieval models: BM25 (Robertson et al., 1995) a statistical baseline, and , MPNet (Song et al., 2020), MiniLM (Wang et al., 2020), and BGE-1.5 (Xiao et al., 2023), which represent SoTA dense retrieval baselines.” (p.6)
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Q103	“SlideVQA (Tanaka et al., 2023) is a multi-image document VQA dataset that contains over 2,600 slide decks, comprising more than 52,000 slide images and 14,500 questions regarding the slide content.” (p.14)
WEB-5	bbOCR outperforms Tesseract on Bengali across all measured domains; Tesseract recall systematically poor on Bengali (https://arxiv.org/pdf/2308.10647)
WEB-8	Cross-lingual Bengali retrieval Recall@1: mE5-large-instruct 27.6%, BGE-M3 21.7%, LaBSE 15.9% — all substantially lower than monolingual settings (https://arxiv.org/pdf/2601.10205)

WEB-11	Bengali document QA requires specialized evaluation; current benchmarks use human-verified ground truth rather than lexical overlap (https://arxiv.org/pdf/2609.03597)

Information Gaps

- BAEC on-premise GPU/compute inventory is not publicly documented
- Proportion of born-digital vs. scanned PDFs in BAEC administrative corpus is unknown

Requires Expert Verification

- BAEC IT infrastructure team confirmation of available compute for on-premise multilingual LLM deployment
- BAEC document management team confirmation of scan quality standards and born-digital proportion

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Remediation

Gap 1: PyTesseract OCR and English-centric embeddings fail on Bengali; cloud LLM dependence incompatible with BAEC air-gap

Recommendation: Evaluate bbOCR [WEB-5] as the Bengali OCR replacement; benchmark multilingual-e5-large-instruct and BGE-M3 [WEB-8, WEB-9] on nuclear-domain Bengali/English/German corpora; deploy exclusively open-weight LLMs (Qwen2-VL-7B or larger) that can run on-premise; verify BAEC compute inventory before finalizing model choice.

ID	Evidence
WEB-5	bbOCR outperforms Tesseract on Bengali across all measured domains; Tesseract recall systematically poor on Bengali (https://arxiv.org/pdf/2308.10647)
WEB-8	Cross-lingual Bengali retrieval Recall@1: mE5-large-instruct 27.6%, BGE-M3 21.7%, LaBSE 15.9% — all substantially lower than monolingual settings (https://arxiv.org/pdf/2601.10205)
WEB-9	mE5-large-instruct is leading multilingual embedding but still exhibits ~25% relative cross-lingual performance drop (https://arxiv.org/pdf/2402.05672)

Output Ontology (OO)

Score: 2/5 — Significant gaps | Confidence: medium

Benchmark: *VISDOMBENCH* | ***Context:*** *Bangladesh Atomic Energy Commission — Internal Document Repository QA*

Output Ontology definition: Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The ground-truth schema is formally flexible — free-text answers with binary/short-text accommodations [Q22, Q64] and arithmetic expressions for numerical answers [Q105]. This formal flexibility is a modest advantage: no fixed categorical ontology is imposed. However, the implicit decision boundaries encoded in the ground-truth labels reflect what counts as 'correct' in CS/ML academic context — correctness is validated against ArXiv paper content, chart values, and slide statements [Q25, Q26]. For BAEC, correctness criteria would involve nuclear safety regulatory thresholds, Bengali administrative terminology, and IAEA/BAERA compliance judgments — none of which are represented. A nuclear regulatory multi-hop QA benchmark [WEB-22] exists precisely because the correctness criteria for regulatory documents (requiring faithful attribution and regulatory judgment) differ from academic QA.

ID	Evidence
Q22	“Every data point in VisDoMBench can be expressed as triple (q, D, aˆ), where a question q is posed to a set of documents D, with groundtruth answer aˆ.” (p.3)
Q25	“Our corpus comprises test/eval sets sourced from several established datasets. We incorporated the PaperTab and FeTaTab splits from the UDA Benchmark(Hui et al., 2024), which in turn sourced these datasets from QASPER(Dasigi et al., 2021) and FeTaQA(Nan et al., 2022), respectively.” (p.3)
Q26	“For chart-based question-answering samples, we drew from SciGraphQA (Li and Tajbakhsh, 2023), which is multi-turn QA dataset on charts from scientific papers, and SPIQA(Pramanick et al., 2024), a chart and table QA dataset system sourced from (Dasigi et al., 2021). Additionally, we included SlideVQA (Tanaka et al., 2023), a multi-image, multi-hop QA dataset centered on presentation slide decks.” (p.3)
Q64	“For PaperTab, we borrow the modified implementation of Word Overlap F1 from (Hui et al., 2024), which takes into account different answer types (binary, short text).” (p.6)
Q105	“It also provides annotated arithmetic expressions for numerical answers, enhancing numerical reasoning capabilities.” (p.14)
WEB-22	Nuclear regulatory QA benchmarks use faithfulness measures (RAGAS Faithfulness 0.93) rather than Word Overlap F1 to assess safety-critical correctness (https://arxiv.org/html/2606.29399v1)

Strengths

- Free-text answer schema [Q22] is flexible enough to accommodate BAEC's diverse answer types without ontology redesign
- Modified Word Overlap F1 for PaperTab [Q64] demonstrates the authors' willingness to adapt metrics for different answer types (binary, short text) — a transferable methodological pattern
- Evidence attribution built into the output (via Evidence Curation step) [Q18, Q53] aligns with the provenance-attribution needs of a nuclear repository

ID	Evidence
Q18	“Moreover, the evidence curation step provides an additional advantage of answer verifiability, since context attribution is built into our approach.” (p.2)
Q22	“Every data point in VisDoMBench can be expressed as triple (q, D, aˆ), where a question q is posed to a set of documents D, with groundtruth answer aˆ.” (p.3)

Q53	“Evidence Curation: As a first step, we prompt the LLM to extract relevant evidence from the retrieved context. The LLM must isolate key sections, such as paragraphs, tables, and figure details, that are most likely to address the query and verbalize them in a structured form. This curation is crucial in a multi-document setup, where non-uniform sources introduce irrelevant, distracting, or adversarial content. Accurately identifying relevant information enhances the model's reasoning abilities by filtering out noise and helps mitigate LLM hallucinations.” (p.5)
Q64	“For PaperTab, we borrow the modified implementation of Word Overlap F1 from (Hui et al., 2024), which takes into account different answer types (binary, short text).” (p.6)

Checklist

Checklist item definitions:

ID	Assessment Question
OO-1	Review output label categories for regional relevance
OO-2	Identify missing regionally specific categories
OO-3	Flag categories encoding non-regional values
OO-4	Consider stakeholder-driven taxonomy redesign
OO-5	Document taxonomy issues harming structural/content validity

Checklist responses:

OO-1: The output label space (free-text, binary, short-text, numerical) [Q22, Q64] is structurally flexible and formally applicable to BAEC answer types. However, the regional relevance of actual label content is zero — ground-truth answers concern CS/ML paper content exclusively [Q100].

OO-2: Missing categories: regulatory compliance verdicts (compliant/non-compliant per BAERA), safety threshold judgments with units and tolerances, Bengali-language free-text answers, and multilingual answer validation (e.g., answer in Bengali referencing an English source document).

OO-3: The ground-truth labels implicitly encode Western academic correctness norms — what a CS graduate reviewer deems a valid answer against an ArXiv paper [Q43, Q112]. This encodes neither nuclear safety domain-expert judgment nor Bengali administrative-document interpretive conventions.

OO-4: Stakeholder-driven taxonomy work would be required: BAEC regulatory officers would need to define correctness criteria for safety-threshold answers; Bengali-literate administrators would need to define correctness for Bengali official-register answers.

OO-5: Structural validity harms: the output schema does not accommodate regulatory judgment or compliance verdicts as distinct answer types. Content validity harms: the actual correctness criteria encoded in ground truth do not reflect BAEC domain expert judgments [WEB-20, WEB-22].

ID	Evidence
Q22	“Every data point in VisDoMBench can be expressed as triple (q, D, aˆ), where a question q is posed to a set of documents D, with groundtruth answer aˆ.” (p.3)
Q43	“Once the augmented queries are generated, a human annotator reviews them using a predefined rubric. The rubric guides the annotator to either select one of the five generated questions, retain the original question, or mark all questions (synthetic and actual) as ambiguous, in which case, the data point is discarded.” (p.5)
Q64	“For PaperTab, we borrow the modified implementation of Word Overlap F1 from (Hui et al., 2024), which takes into account different answer types (binary, short text).” (p.6)
Q100	“The dataset was constructed from 290,000 Computer Science and Machine Learning papers published on ArXiv between 2010 and 2020, with the help of Palm-2 generating 295,000 samples of open-vocabulary multi-turn question-answer dialogues about the graphs.” (p.14)
Q112	“The reviewer is a graduate student who is paid at the hourly rate for Graduate Assistants at the university where they are a student.” (p.15)
WEB-20	NuclearQA was expert-authored (100 questions) — nuclear-domain correctness requires expert annotation (https://arxiv.org/pdf/2310.10920)

WEB-22	Nuclear regulatory QA benchmarks use faithfulness measures (RAGAS Faithfulness 0.93) rather than Word Overlap F1 to assess safety-critical correctness (https://arxiv.org/html/2606.29399v1)

Information Gaps

- BAEC's own definition of 'correct answer' for safety-critical queries (e.g., tolerance ranges around threshold values) is not documented

Requires Expert Verification

- BAEC regulatory officers and safety engineers would need to specify correctness criteria and tolerance ranges for safety-threshold answers

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Remediation

Gap 1: Output schema does not distinguish regulatory compliance verdicts, safety-threshold answers with tolerances, or multilingual answer validation

Recommendation: Extend the (q, D, â) schema with answer-type annotations for regulatory judgment, numerical thresholds with unit/tolerance metadata, and source-language tagging; involve BAERA regulatory officers in defining correctness criteria for safety-critical answer types.

Output Content (OC)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: *VISDOMBENCH* | ***Context:*** *Bangladesh Atomic Energy Commission — Internal Document Repository QA*

Output Content definition: Whether ground-truth labels align with the judgments of regional stakeholders.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Annotation is performed by a single graduate student reviewer [Q43, Q111, Q112] with a predefined rubric for selecting among GPT-4o-generated query variants [Q113]. Annotator identity is kept confidential and no PII is reported [Q86]; linguistic background, domain expertise, and demographic composition are not disclosed. No inter-annotator agreement is reported since only one annotator is used. For BAEC, the ground-truth labels have zero convergent validity: no Bengali-proficient annotator, no German/Russian-proficient annotator, no nuclear science domain expert, no BAEC staff involvement, and no IAEA/BAERA regulatory-literate reviewer participated in label creation. Bengali legal document QA has been shown to require expert annotator pools [WEB-11], and nuclear-domain correctness requires expert judgment [WEB-20] — neither condition is met by VisDoMBench's single-reviewer protocol.

ID	Evidence
Q43	“Once the augmented queries are generated, a human annotator reviews them using a predefined rubric. The rubric guides the annotator to either select one of the five generated questions, retain the original question, or mark all questions (synthetic and actual) as ambiguous, in which case, the data point is discarded.” (p.5)
Q86	“The identities of human evaluators remain confidential, and no personally identifiable information (PII) is used at any stage of our experiments.” (p.9)
Q111	“We employ a human reviewer to assess the quality of the generated queries and select one of the queries or reject all queries.” (p.15)
Q112	“The reviewer is a graduate student who is paid at the hourly rate for Graduate Assistants at the university where they are a student.” (p.15)
Q113	“Fig 19 gives a brief of the instructions as well as the evaluation rubric given to the reviewer.” (p.15)
WEB-11	Bengali document QA requires specialized evaluation; current benchmarks use human-verified ground truth rather than lexical overlap (https://arxiv.org/pdf/2609.03597)
WEB-20	NuclearQA was expert-authored (100 questions) — nuclear-domain correctness requires expert annotation (https://arxiv.org/pdf/2310.10920)

Strengths

- The rubric-driven annotation protocol [Q43, Q44, Q45, Q113] is methodologically disciplined — the annotator cross-references localized evidence and performs a document-collection ambiguity check — and this methodology is transferable if re-executed with appropriate expert annotators
- Explicit documentation of the annotator role, compensation, and task [Q111, Q112] provides transparency that aids reproducibility

ID	Evidence
Q43	“Once the augmented queries are generated, a human annotator reviews them using a predefined rubric. The rubric guides the annotator to either select one of the five generated questions, retain the original question, or mark all questions (synthetic and actual) as ambiguous, in which case, the data point is discarded.” (p.5)
Q44	“The annotator is tasked with ensuring that the question is sufficiently specific by cross-referencing the localized evidence.” (p.5)
Q45	“Additionally, the annotator performs a simple search across the entire document collection to verify that the question cannot be ambiguously answered by any other document.” (p.5)

Q111	“We employ a human reviewer to assess the quality of the generated queries and select one of the queries or reject all queries.” (p.15)
Q112	“The reviewer is a graduate student who is paid at the hourly rate for Graduate Assistants at the university where they are a student.” (p.15)
Q113	“Fig 19 gives a brief of the instructions as well as the evaluation rubric given to the reviewer.” (p.15)

Checklist

Checklist item definitions:

ID	Assessment Question
OC-1	Do ground-truth labels reflect regional stakeholder perspectives?
OC-2	Assess potential annotator-regional population disagreement
OC-3	Review annotator demographics from documentation
OC-4	Consider label re-annotation by regional annotators
OC-5	Review aggregation methods for minority perspective erasure
OC-6	Document label issues harming convergent/external validity

Checklist responses:

OC-1: No — ground-truth labels reflect a single CS graduate student's judgment [Q112] over CS/ML academic content, not BAEC stakeholder perspectives.

OC-2: High potential for disagreement: BAEC regulatory officers and nuclear engineers would evaluate correctness against IAEA/BAERA standards and reactor safety considerations rather than against ArXiv paper content. The capability gap evidenced by NuclearQA [WEB-20] and KhatianDoc [WEB-11] suggests substantial domain-expert vs. non-expert disagreement.

OC-3: Annotator demographics are not disclosed beyond 'graduate student...paid at the hourly rate for Graduate Assistants' [Q112]; identity is kept confidential [Q86]. No Datasheet or Data Statement specifying language proficiency, domain expertise, or demographic representativeness is provided.

OC-4: Full re-annotation by a BAEC-representative annotator pool (nuclear scientists, Bengali-literate administrators, BAERA regulatory officers) would be required — the benchmark provides no coverage of this.

OC-5: Aggregation is not applicable since only a single annotator reviews each item [Q43]; this itself erases any possibility of capturing minority perspectives.

OC-6: Convergent validity harms: single-annotator labels cannot be validated against regional stakeholder judgments. External validity harms: labels do not generalize to a nuclear repository context [WEB-20, WEB-22].

ID	Evidence
Q43	“Once the augmented queries are generated, a human annotator reviews them using a predefined rubric. The rubric guides the annotator to either select one of the five generated questions, retain the original question, or mark all questions (synthetic and actual) as ambiguous, in which case, the data point is discarded.” (p.5)
Q86	“The identities of human evaluators remain confidential, and no personally identifiable information (PII) is used at any stage of our experiments.” (p.9)
Q112	“The reviewer is a graduate student who is paid at the hourly rate for Graduate Assistants at the university where they are a student.” (p.15)
WEB-11	Bengali document QA requires specialized evaluation; current benchmarks use human-verified ground truth rather than lexical overlap (https://arxiv.org/pdf/2609.03597)
WEB-20	NuclearQA was expert-authored (100 questions) — nuclear-domain correctness requires expert annotation (https://arxiv.org/pdf/2310.10920)
WEB-22	Nuclear regulatory QA benchmarks use faithfulness measures (RAGAS Faithfulness 0.93) rather than Word Overlap F1 to assess safety-critical correctness (https://arxiv.org/html/2606.29399v1)

Information Gaps

- Annotator's exact linguistic and domain profile is not disclosed beyond 'graduate student'
- No inter-annotator agreement data exist to characterize annotation reliability

Requires Expert Verification

- A BAEC-representative annotator pool (Bengali-proficient administrators, nuclear scientists, BAERA regulatory officers) would need to be convened to re-annotate any adapted benchmark instances

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Remediation

Gap 1: Single CS graduate annotator provides no convergent validity with BAEC domain expert judgment

Recommendation: Convene a multi-annotator BAEC pool including Bengali-proficient administrative staff, nuclear engineers, and BAERA regulatory officers; measure inter-annotator agreement; follow the KhatianDoc [WEB-11] human-verification protocol for Bengali and NuclearQA [WEB-20] expert-authored protocol for nuclear items.

ID	Evidence
WEB-11	Bengali document QA requires specialized evaluation; current benchmarks use human-verified ground truth rather than lexical overlap (https://arxiv.org/pdf/2609.03597)
WEB-20	NuclearQA was expert-authored (100 questions) — nuclear-domain correctness requires expert annotation (https://arxiv.org/pdf/2310.10920)

Output Form (OF)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: *VISDOMBENCH* | ***Context:*** *Bangladesh Atomic Energy Commission — Internal Document Repository QA*

Output Form definition: Whether the expected output modality matches regional deployment needs and accessibility.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

VisDoMBench evaluates outputs via Word Overlap F1 [Q65] with a PaperTab-specific variant [Q64], plus ANLCS for evidence retrieval [Q58, Q66] and document-identification rate [Q59, Q60]. Word Overlap F1 is a language-sensitive lexical-overlap metric that computes token-level similarity between prediction and reference. This metric is invalid for Bengali — token boundaries, conjunct consonants, and morphology differ from English — and problematic for German compound words. The output modality (free-text) matches BAEC's needs, but the scoring function cannot be applied validly to multilingual outputs. The benchmark does not evaluate text-to-speech or any accessibility output modality; however, this is not a specified requirement for BAEC's deployment. Multiple-LLM-call cost and hallucination risk are acknowledged limitations [Q89, Q90], and the latter is particularly salient for safety-critical nuclear outputs.

ID	Evidence
Q58	“Evidence extraction is assessed using ANLCS between ground truth evidence and retrieved chunks/pages.” (p.6)
Q59	“Document identification evaluates the retrievers' ability to select the correct source document in a multi-document setup.” (p.6)
Q60	“We report the rate of instances where the ground truth document is the source of the majority of the retrieved context.” (p.6)
Q64	“For PaperTab, we borrow the modified implementation of Word Overlap F1 from (Hui et al., 2024), which takes into account different answer types (binary, short text).” (p.6)
Q65	“For all other datasets, we report the Word Overlap F1, which serves as a flexible metric to evaluate different answer types.” (p.6)
Q66	“Fig. 3 presents the performance of various retrieval models in extracting evidence from documents, evaluated using the Averaged Normalized Longest Common Subsequence (ANLCS) between retrieved evidence and ground truth evidence, for different context window lengths (k = [1, 5, 10, 20]).” (p.7)
Q89	“(2) Multiple LLM calls: Our methodology necessitates multiple LLM calls; specifically, we make three LLM calls per query. While this approach may not be optimal, it is still more cost-effective than utilizing long-context models.” (p.9)
Q90	“(3) Hallucinations: As with all works involving large language models (LLMs), our approach is subject to inherent limitations related to AI safety and the risk of hallucination. These issues can affect the reliability and accuracy of the generated outputs and underscore safety risks, highlighting the need for ongoing research and refinement in the field of AI to mitigate these challenges.” (p.9)

Strengths

- Free-text output modality [Q22] matches the modality BAEC staff would receive from a repository QA system
- ANLCS-based evidence-retrieval scoring [Q58, Q66] and document-identification rate [Q60] are provenance-aware metrics that align with BAEC's need for traceable attribution
- Explicit acknowledgment of hallucination risk as a limitation [Q90] mirrors BAEC's safety-critical reliability concern

ID	Evidence
Q22	“Every data point in VisDoMBench can be expressed as triple (q, D, aˆ), where a question q is posed to a set of documents D, with groundtruth answer aˆ.” (p.3)
Q58	“Evidence extraction is assessed using ANLCS between ground truth evidence and retrieved chunks/pages.” (p.6)
Q60	“We report the rate of instances where the ground truth document is the source of the majority of the retrieved context.” (p.6)

Q66	“Fig. 3 presents the performance of various retrieval models in extracting evidence from documents, evaluated using the Averaged Normalized Longest Common Subsequence (ANLCS) between retrieved evidence and ground truth evidence, for different context window lengths (k = [1, 5, 10, 20]).” (p.7)
Q90	“(3) Hallucinations: As with all works involving large language models (LLMs), our approach is subject to inherent limitations related to AI safety and the risk of hallucination. These issues can affect the reliability and accuracy of the generated outputs and underscore safety risks, highlighting the need for ongoing research and refinement in the field of AI to mitigate these challenges.” (p.9)

Checklist

Checklist item definitions:

ID	Assessment Question
OF-1	Does output modality match regional deployment needs?
OF-2	Assess text-to-speech availability for speech output
OF-3	Consider literacy rates and accessibility requirements
OF-4	Document form mismatches harming external validity

Checklist responses:

OF-1: Output modality (free-text answers) matches BAEC needs. However, the scoring metric (Word Overlap F1) is a chunk/token overlap measure built on English tokenization assumptions [Q65] — invalid for Bengali script and poorly suited to German compound nouns.

OF-2: Text-to-speech is not evaluated by VisDoMBench. The BAEC deployment context does not indicate TTS as a requirement, so this is not a documented mismatch for this deployment.

OF-3: BAEC users are described as educated professionals (scientists, engineers, regulatory officers, administrators) with high occupational literacy; text-output modality is appropriate. No specific accessibility requirements are documented in the deployment context.

OF-4: Form mismatches harming external validity: (a) Word Overlap F1 cannot validly score Bengali-script answers [Q65]; (b) the metric would systematically mis-score German compound-noun answers; (c) no multilingual evaluation metric is proposed or discussed [Q64, Q65]; (d) safety-critical hallucination mitigation is acknowledged but not instrumented [Q90].

ID	Evidence
Q22	“Every data point in VisDoMBench can be expressed as triple (q, D, aˆ), where a question q is posed to a set of documents D, with groundtruth answer aˆ.” (p.3)
Q64	“For PaperTab, we borrow the modified implementation of Word Overlap F1 from (Hui et al., 2024), which takes into account different answer types (binary, short text).” (p.6)
Q65	“For all other datasets, we report the Word Overlap F1, which serves as a flexible metric to evaluate different answer types.” (p.6)
Q90	“(3) Hallucinations: As with all works involving large language models (LLMs), our approach is subject to inherent limitations related to AI safety and the risk of hallucination. These issues can affect the reliability and accuracy of the generated outputs and underscore safety risks, highlighting the need for ongoing research and refinement in the field of AI to mitigate these challenges.” (p.9)
WEB-11	Bengali document QA requires specialized evaluation; current benchmarks use human-verified ground truth rather than lexical overlap (https://arxiv.org/pdf/2609.03597)
WEB-22	Nuclear regulatory QA benchmarks use faithfulness measures (RAGAS Faithfulness 0.93) rather than Word Overlap F1 to assess safety-critical correctness (https://arxiv.org/html/2606.29399v1)

Information Gaps

- No empirical study of Word Overlap F1 behavior on Bengali-script or German compound-noun answers
- No safety-critical output reliability thresholds specified for BAEC's use case

Requires Expert Verification

- BAEC safety engineers would need to specify acceptable hallucination and incorrect-answer rates for safety-threshold queries

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Remediation

Gap 1: Word Overlap F1 is invalid for Bengali and poorly suited to German compound nouns; no faithfulness/hallucination instrumentation for safety-critical outputs

Recommendation: Replace Word Overlap F1 with script-aware tokenization-adjusted metrics for Bengali and semantic-similarity metrics for cross-lingual answers; add RAGAS Faithfulness or equivalent attribution-faithfulness measure [WEB-22] for safety-critical queries; establish per-query-category reliability thresholds with BAEC safety engineers.

ID	Evidence
WEB-22	Nuclear regulatory QA benchmarks use faithfulness measures (RAGAS Faithfulness 0.93) rather than Word Overlap F1 to assess safety-critical correctness (https://arxiv.org/html/2606.29399v1)